

UEC714: FIBER OPTIC COMMUNICATION

L	T	P	Cr
2	0	2	3.0

Course Objectives: To understand fiber optic communication system, transmitter section, medium- the optical fiber, receiver section, analyze system based on important parameters for characterizing optical fiber, optical source, detector and amplifier, fundamentals and advances in lasers, LEDs, photodiodes.

Optical Fibers and Their Characteristics: Introduction to high frequency communication, nature of light, advantages of optical communication, fiber structures, wave guiding, basic optical laws and definitions, optical fiber modes and configuration, mode theory for circular waveguides, single mode fibers, graded index fiber, fiber materials, fabrication and mechanical properties, fiber optic cables; joints, splices, connectors, attenuation, signal distortion, dispersion and polarization mode dispersion in optical fibers, mode coupling, design optimization of single mode fibers.

Optical Sources and Amplifiers: Light emitting diodes, semiconductor laser, various configurations of semiconductor laser, performance parameters of LEDs and semiconductor lasers, light source linearity, modal partition and reflection noise, reliability consideration; power launching and coupling, optical amplifiers: erbium doped fiber amplifier, semiconductor optical amplifier, Raman amplifier.

Photodetectors: Operating principle and physical properties of photodiodes, pin and avalanche photodiodes, photodetector noise, response time, avalanche multiplication noise, temperature effect on avalanche gain, photodiode material.

Optical Communication Systems: Optical receiver operation- fundamental receiver operation, digital receiver performance calculation, preamplifier types, analog receivers. digital transmission systems- point to point links, line coding, noise effects on system performance. analog system: overview of analog links, carrier to noise ratio, multichannel transmission techniques, WDM: basics and components, LAN, coherent optical fiber communication- classification of coherent system, requirements on semiconductor lasers, modulation techniques.

Familiarization with standards: *IEEE 404-1977, IEEE 812-1984, ITU G.651-G.657, ANSI Z136.2.*

LABORATORY WORK: Basic optical communication link experiments (analog & digital), measurement of numerical aperture, splicing, multiplexing experiments, bending losses, measurement with OTDR, design and performance analysis using simulation tools.

Course Learning Outcomes (CLOs):

The students will be able to:

1. Identify and formulate the types, basic properties and transmission characteristics of optical fibers.
2. Analyse different types of optical sources and amplifiers for efficient optical fiber communication.
3. Analyse and formulate pin and avalanche photodetectors in optical fiber communication systems.

4. Realize the analog and digital fiber optic communication systems and networks with different modulation techniques.

Test Books:

1. Keiser, Gred, Optical Fiber Communications, Tata McGraw-Hill, (2013) 5th ed.
2. Senior, John M., and Yousif Jamro, M., Optical fiber communications: principles and practice, Prentice Hall, (2009) 3rd ed.

Reference Books:

1. Ajoy Kumar Ghatak and K. Thyagarajan, Optical Electronics, Cambridge University Press (2012) 2nded.
2. Bahaa E. A. Saleh, Malvin C. Teich, Fundamentals of Photonics, John Wiley & Sons, (2019) 3rd ed.